EE1007
Introduction to Computer Science Laboratory

Lab 8
Array

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Lab 8.1
Average Score

- You are asked to write a program to calculate the average of three scores and find the fail ones.
- This program will ask the user to enter three integer for the scores. The range of score is [0, 100].
- After that, the program will print the average score with three decimal places.
- If there are score(s) less than 60 points, please display the word “fail:”, following by the order and the failed score(s) in the sequence.
- You need to use an array to store the scores and calculate the average of the scores. Otherwise, you will get 0 point.

Lab 8.1
Average Score

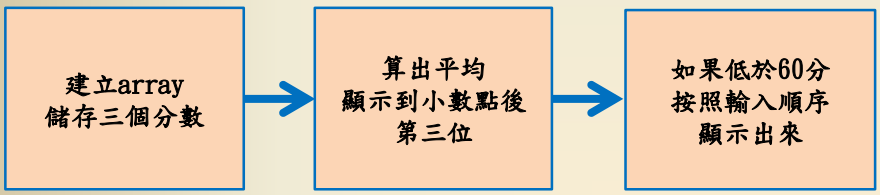
◆The screen dialog should appear as follows:

```
Please enter the scores of three students:
30
50
80
avg = 53.333
fail:
1: 30
2: 50

Process returned 0 (0x0)   execution time : 9.100 s
Press any key to continue.
```


Lab 8.1
Average Score

◆思考方向：



Lab 8.1
Average Score

◆Step 1. 建立array，儲存三個分數

```
int score[3];
int sum = 0;
double avg;
for(i=0;i<3;i++)
{
   cin>>score[i];
   sum = sum+score[i];
}
```


Lab 8.1
Average Score

◆Step 2. 算出平均，顯示小數點後第三位

```
avg = (double)sum/3;
cout<<"avg = "<<fixed<<setprecision(3)<<avg<<endl;
```

Lab 8.1
Average Score

◆Step 3. 如果低於60分，按照輸入順序顯示出來

```
for(i=0;i<3;i++)
{
   if(score[i]<60)
   {
      cout<<"fail:\n";
      break;
   }
}

for(i=0;i<3;i++)
{
   if(score[i]<60)
   {
      cout<<(i+1)<<": "<<score[i]<<endl;
   }
}
```


Lab 8.2
Arithmetic Sequence

◆You are asked to write a program to show the Arithmetic Sequence

◆This program will ask the user to enter the three integer for the first term(*a1*), common difference(*d*) and the number of terms(*n*).

◆The range of the integer is $[-2^{31}-1, 2^{31}-1]$.

◆The program will display the arithmetic series as a1+a2+…+an = sum.

◆You must use array to store all terms.

◆If the term is an negative integer, it should include parentheses in the above equation as (-a1)+(-a2)+…(-an) = (-sum).

Lab 8.2
Arithmetic Sequence

◆The screen dialog should appear as follows:

```
Please enter the first term(a1), the common difference(d) and the number of terms(n):
1
4
5
The arithmetical series: 1 + 5 + 9 + 13 + 17 = 45
Process returned 0 (0x0)   execution time : 6.968 s
Press any key to continue.
```


Lab 8.2
Arithmetic Sequence

◆思考方向：



Lab 8.2
Arithmetic Sequence

◆Step 1&2. 建立array，儲存等差數列

```
cin>>a1>>d>>n;
int ary[100];
int i;
cout<<"The arithmetical series: ";
for(i=0;i<n;i++)
{
    ary[0]=a1;
    ary[i+1]=ary[i]+d;
    sum=sum+ary[i];
```


Lab 8.2
Arithmetic Sequence

◆Step 3. 輸出相加後的結果

```
if(i==n-1)
{
   if(ary[i]<0)
   {
      cout<<"("<<ary[i]<<") = ";
      if(sum<0)
         cout<<"("<<sum<<")";
      else
         cout<<sum;
   }
   else
   {
      cout<<ary[i]<<" = ";
      if(sum<0)
         cout<<"("<<sum<<")";
      else
         cout<<sum;
   }
}
else
{
   if(ary[i]<0)
      cout<<"("<<ary[i]<<") + ";
   else
      cout<<ary[i]<<" + ";
}
```

Lab 8.3
Pascal's Triangle

- You are asked to write a program to print Pascal's Triangle based on a given integer
- This program will ask the user to enter a positive integer n. After that, the program will display Pascal's triangle with *n* columns.
- Then the program will ask the user to enter a positive integer again, and print the corresponding Pascal's triangle with *n* columns .
- The program will terminate until the user enters 0.
- The range if n is [1,30]
- You need to store the values of Pascal's triangle with a multidimension array.

Lab 8.3
Pascal's Triangle

- The screen dialog should appear as follows:

```
Please enter a number for the column of the Pascal triangle:
4
1
1  1
1  2  1
1  3  3  1

Please enter a number for the column of the Pascal triangle:
5
1
1  1
1  2  1
1  3  3  1
1  4  6  4  1

Please enter a number for the column of the Pascal triangle:
8
1
1  1
1  2  1
1  3  3  1
1  4  6  4  1
1  5  10  10  5  1
1  6  15  20  15  6  1
1  7  21  35  35  21  7  1

Please enter a number for the column of the Pascal triangle:
0

Process returned 0 (0x0)   execution time : 18.095 s
Press any key to continue.
```

Lab 8.3
Pascal's Triangle

◆思考方向：



Lab 8.3
Pascal's Triangle

◆Step 1&2. 建立二維陣列，存入Pascal's triangle

```
const int arraysize = 34;
int A [arraysize] [arraysize];

for ( int i = 0; i < arraysize; i++ )
{
   A [i][0] = 1;
   A [i][i] = 1;
}

for ( int i = 2; i < arraysize; i++ )
{
   for ( int j = 1; j < i; j++ )
      A [i][j] = A [i-1][j-1] + A [i-1][j];
}
```

Lab 8.3
Pascal's Triangle

◆Step 3.顯示Pascal's triangle

```
while(cin>>n && n !=0 )
{
   for (int i=0; i<n; i++)
   {
      for ( int j = 0; j <= i; j++ )
         cout << A[i][j] << "  ";
      cout << endl;
   }
   cout << endl;
```


Lab 8.4
NCU Beverage Ordering System

◆As a start-up company, the NCU Beverage only has three types of beverage in the ordering system: cold drink, coffee and hot drink, respectively.

◆To provide rich choices, both cold drink and coffee consist of **eight** different kinds of taste, and hot drink consists of **four** different kinds of taste.

Lab 8.4
NCU Beverage Ordering System





- -----Menu-----
  Cold Drink
  1. Coca Cola 40
  2. Coca Cola Zero 40
  3. Lemon Tea 55
  4. Green Tea 55
  5. Black Tea 60
  6. Milk Tea 60
  7. Apple Juice 55
  8. Berry Juice 55

  Coffee
  1. Espresso 55
  2. Hot Americano 55
  3. Iced Americano 60
  4. Hot Latte 70
  5. Iced Latte 85
  6. Toffee Latte 95
  7. Rose Latte 100
  8. Iced Rose Latte 110

  Hot Drink
  1. Hot Chocolate 85
  2. Chocolate Milk 100
  3. Matcha Milk 100
  4. Rose Woolong 100



Lab 8.4
NCU Beverage Ordering System





- The program will ask the user to input which kind of drink he/she wants. 1 for cold drink, 2 for coffee and 3 for hot drink.
- After the user inputs which kind of drink he/she wants, the program will ask the user to input which kind of taste he/she wants. (EX:1~4, 1~8)
- And the program will execute from the first action again until the user input 0 to end the system.
- Finally, the program will show all the drink and the price the user has bought.



Lab 8.4
NCU Beverage Ordering System

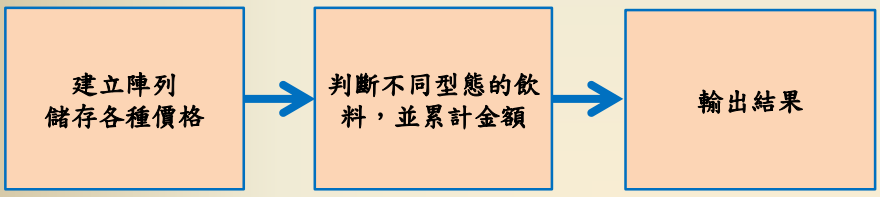
◆The screen dialog should appear as follows:

```
-----Menu-----
1
2
3
2
0
Total Cost: 140

Process returned 0 (0x0)   execution time : 6.617 s
Press any key to continue.
```


Lab 8.4
NCU Beverage Ordering System

◆思考方向：



Lab 8.4
NCU Beverage Ordering System

◆Step 1. 建立array，儲存各種價格

```
const int coldnumber  = 8, coffeenumber = 8;
const int hotnumber = 4;

int coldprice[coldnumber] = {40, 40, 55, 55, 60, 60, 55, 55};
int coffeeprice[coffeenumber] = {55, 55, 60, 70, 85, 95, 100, 110};
int hotprice[hotnumber] = {85, 100, 100, 100};
```


Lab 8.4
NCU Beverage Ordering System

◆Step 2&3. 判斷不同型態的飲料，累計和顯示金額

```
while (cin >> n && n != 0)
{
   int num = 0;
   if (n == 1)
   {
      cin >> num;
      total = total + coldprice[num-1];

   }
   else if (n == 2)
   {
      cin >> num;
      total = total + coffeeprice[num-1];
   }
   else if (n == 3)
   {
      cin>>num;
      total = total + hotprice[num-1];
   }
   else
      cout<<"Please enter again."<<endl;

}
cout << "Total Cost: " << total<<endl;
```

Evaluation

- Lab8.1(30%)
  - 5 testcases, 4% each
- Lab8.2(30%)
  - 5 testcases, 4% each
- Lab8.3(30%)
  - 5 testcases, 4% each
- Lab8.4(20%)
  - 5 testcases, 4% each





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